MILITARY STANDARD

MIL-STD-1320-175C (NAVY)

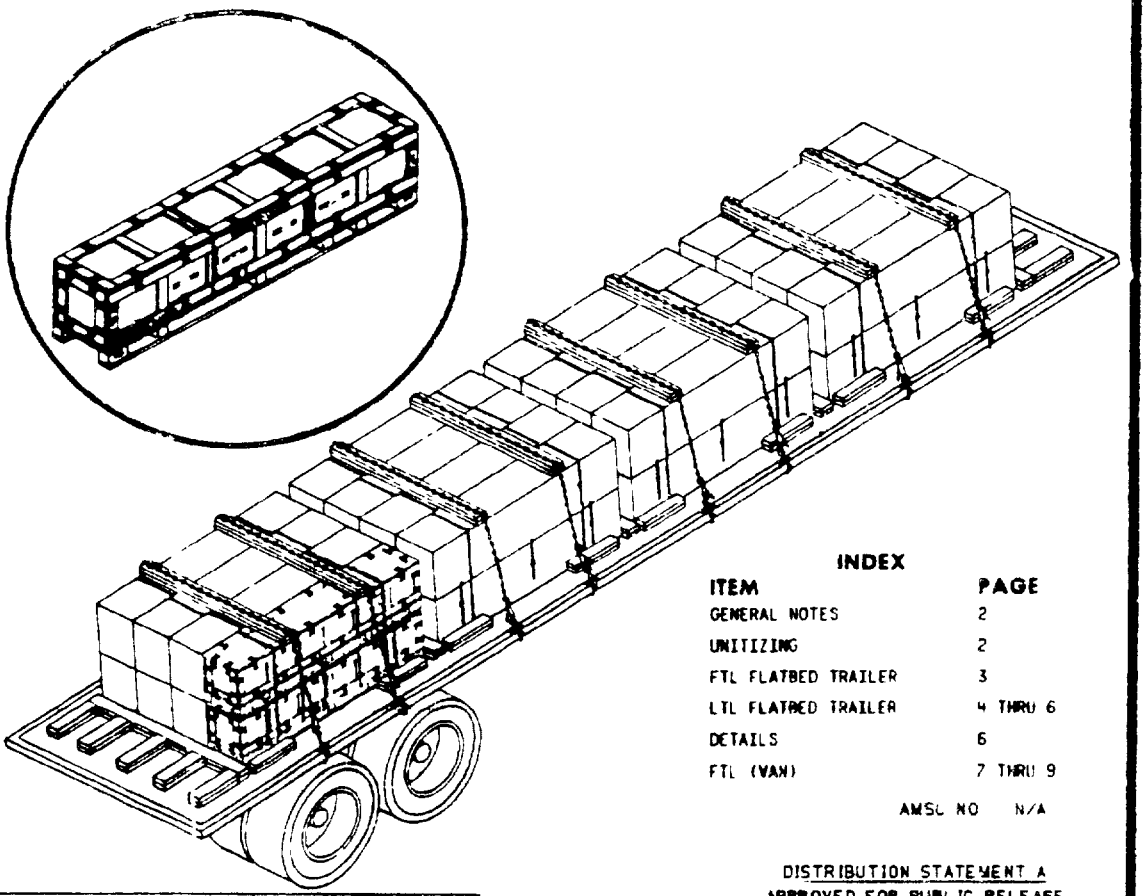
1 JUL 1988

SUPERSEDING MIL-STD-1320-175B (NAVY) 10 MARCH 1987

TRUCKLOADING
ROCKET MOTOR, MK 27 OR MK 56 AND MODS (OTRM) IN CONTAINER, MK 593 MOD 0 AND SUSTAINER, MK 30 MODS IN CONTAINER, MK 737 MOD 0

CONTAINER DATA

DIMENSIONS	110 L X 20 W X 25 H
GROSS WEIGHT	SEE TABLE I
CUBE	31.8 CU FT
DOT HAZARD CLASSIFICATION	EXPLOSIVE B



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FTL (VAN)	7 THRU 9

AMSC NO N/A

DISTRIBUTION STATEMENT A
APPROVED FOR PUBLIC RELEASE
DISTRIBUTION IS UNLIMITED

REV LTR	REVISION DESCRIPTION	DATE	TDA	SYSCOM
			APPROVAL	
C	ADDED MK 30 SUSTAINER	7-6 88	ER	AIC
B	ADDED LONGER TRAILERS			
A	CHANGED TEMP CONT VANS			

NOTES
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING CONTAINERLOADING AND CARLOADING MILITARY STANDARDS REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY)

FSC 8140

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SIGNATURE TCA — DATE 7-6-88

SIGNATURE SEA SYSCOM BY DIRECTION — DATE 7-11-88

ORIGINATOR — 11/3/88 DATE

WPNSTA EARLE, NEW JERSEY

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GENERAL NOTES

1 THIS DOCUMENT PROVIDES DETAILED INSTRUCTION FOR TRUCKLOADING THE DUAL THRUST ROCKET MOTOR (DTRM) MK 27 OR MK 56 MOD MODS IN CONTAINER MK 593 MOD 0 AND THE MK 30 MODS 1 2 AND 4 SUSTAINER IN THE MK 737 MOD 0 CONTAINER FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT MIL-STD-1320 (NAVY)

2 THE PROCEDURES DESCRIBED HEREIN ARE INTENDED FOR 44 FT AND 45 FT FLATBED TRAILERS WITH THE TRAILERS TANDEM AXLES LOCATED IN THE "WESTERN POSITION" (AT THE EXTREME REAR OF THE TRAILER) DO NOT USE TRAILERS WITH METAL FLOORS

3 CHAINS AND STEEL STRAPPING MAY BE USED INTERCHANGEABLY ON A ONE-TO-ONE BASIS FOR TIE DOWNS

4 STEEL STRAPPING USED AS TIE DOWNS SHALL BE 2 X .050 AND SHALL CONFORM TO AND BE APPLIED AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT MIL-STD-1320 (NAVY)

5 CHAINS FITTINGS AND LOAD BINDERS SHALL CONFORM TO AND BE APPLIED AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT MIL-STD-1320 (NAVY) EXCEPT THAT THE CHAIN/GRAB HOOKS SHALL BE ATTACHED TO THE STAKE POCKETS (NOT AROUND THE RUB RAIL)

6 PROTECTOR BOARDS (DOUBLED 2 X 6'S) SHALL BE USED UNDER THE TIE DOWNS WHEN USING STEEL STRAPPING THE STRAPS SHALL BE STAPLED TO THE PROTECTOR BOARD USING 2 INCH STRAPPING STAPLES WHEN USING CHAIN THE CHAINS SHALL BE SECURED TO THE PROTECTOR BOARD BY DRIVING A 10d NAIL THROUGH THE CHAIN LINK AND CLINCHING THE NAIL OVER THE CHAIN. FIVE STAPLES/NAILS SHALL BE USED FOR EACH TIE DOWN

7 PRIOR TO LOADING THE TRAILER AND DURING PRELOADING INSPECTION REQUIRED BY NAVSEA OP 2165 AND REPORTED ON DD FORM 626 THE CHAINS FITTINGS AND LOAD BINDERS SHALL BE INSPECTED FOR STRETCH GOUGING BENT LINKS WEAR AND ANY OTHER NOTICEABLE DEFECTS THE INSPECTOR SHALL CONFIRM THAT THE CHAINS, FITTINGS AND LOAD BINDERS HAVE BEEN INSPECTED AND SHALL SO NOTE IN ITEM NO 22 OF DD FORM 626 ANY DEFICIENCY SHALL BE CAUSE FOR REJECTION OF A CHAIN OR LOAD BINDER

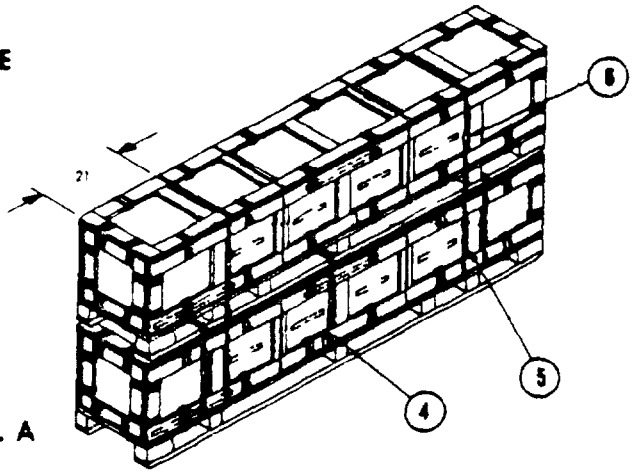
8 UNLESS OTHERWISE SPECIFIED, DUNNAGE LUMBER STEEL STRAPPING, STRAP SEALS AND NAILS SHALL BE AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT MIL-STD-1320 (NAVY)

9 THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS

10 AFTER THE BLOCKING, BRACING AND TIE DOWNS HAVE BEEN INSPECTED, THE CONTAINERS SHALL BE COMPLETELY COVERED WITH FIRE-RESISTANT, WATERPROOF TARPAULINS

11 AFTER THE TARPAULINS ARE IN PLACE, ATTACH "EXPLOSIVES B" PLACARDS TO BOTH SIDES, FRONT AND BACK OF THE TRAILER (WHEN APPLICABLE) AND ATTACH THE SHIPPING DOCUMENTS TO AN ACCESSIBLE AREA ON THE BACK DECK OF THE TRAILER.

UNITIZING PROCEDURE



DETAIL A

1 USING A FORKLIFT TRUCK, STACK TWO CONTAINERS TOGETHER AS SHOWN ABOVE. ORIENT THE TOP CONTAINER OPPOSITE THE BOTTOM CONTAINER (AFT END OF THE TOP CONTAINER OVER THE FORWARD END OF THE BOTTOM CONTAINER)

2 SECURE BOTTOM CONTAINER TO THE TOP CONTAINER WITH FIVE 1-1/4 X .035 STRAPS POSITIONED AS SHOWN IN DETAIL A

3 TENSION AND DOUBLE NOTCH THE SEALS

WARNING

TO AVOID TOPPLING SPECIAL CARE SHOULD BE TAKEN WHEN MOVING STACKS OF CONTAINERS, ESPECIALLY WHEN THE SURFACE IN THE DIRECTION OF THE MOVE IS UNEVEN OR NOT LEVEL THE UNITIZING OF THE CONTAINERS IS ONLY AUTHORIZED AS PART OF THE TRUCKLOADING PROCEDURE. MOVEMENT OF UNITIZED CONTAINERS SHALL BE LIMITED TO THAT NECESSARY TO LOAD OR UNLOAD THE TRAILER THE CENTER OF GRAVITY OF CONTAINERS LOADED WITH THE ROCKET MOTORS IS TOWARDS THE AFT END OF THE CONTAINER WHEN HANDLING THE LOADED CONTAINER(S) WITH A FORKLIFT TRUCK, EXERCISE APPROPRIATE CAUTION

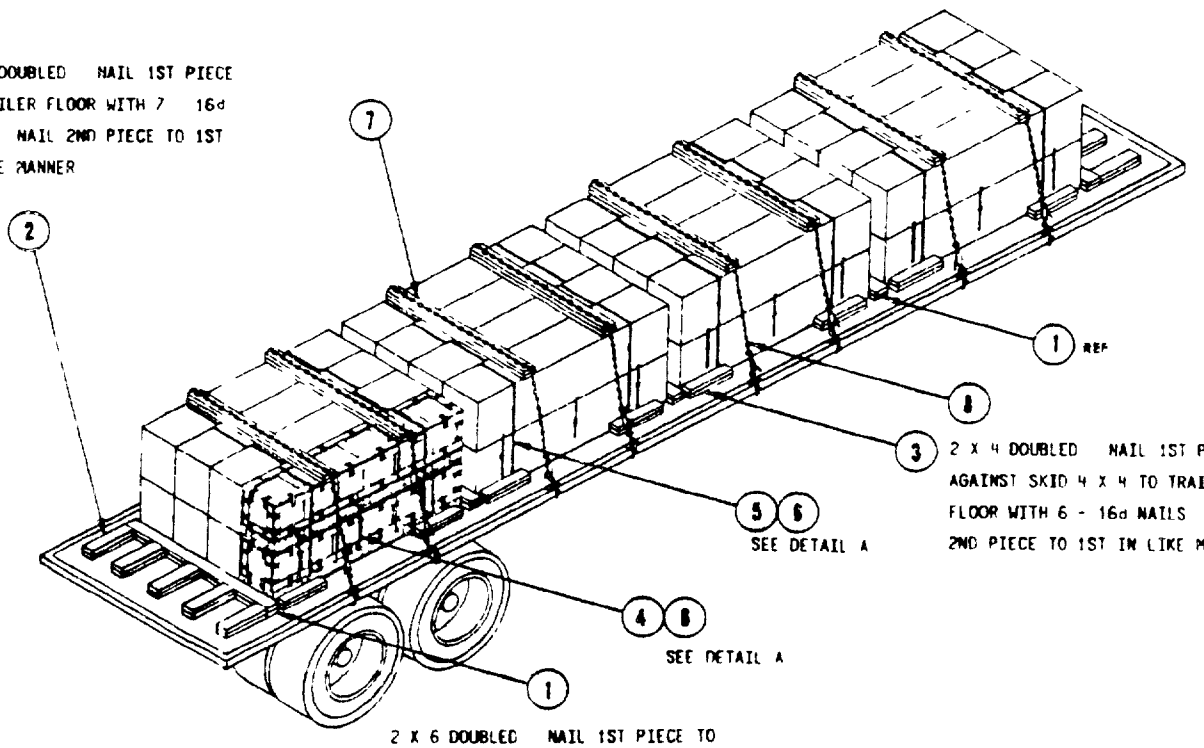
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FTL 44 ST FOR 45 FT FLATBED TRAILER

NOTE
TARPAULIN OVER
CONTAINERS OMITTED
FOR CLARITY

TABLE I

DODIC/NALC	ITEM	GROSS WT (LBS)	FTL WT (LBS)
V178 V279 V289	DTRM MK 2	992	31 744
W436 V878 W451	DTRM MK 56	1182	37 824
V294 V297 V923	SUSTAINERS MK 30 MOD5	825	26 400



REQ'D	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
8	8	CHAIN & LOAD BINDER	NOTES 3, 4 5 & 6	3/8 OR 5/16
16	7	PROTECTOR BOARD	WOOD NOTE 8	2 X 6 X 80
80	6	SEAL	NOTE 8	1-1/4 X .035
32	5	LONG STRAP	STEEL NOTE 8	1-1/4 X .035 X 150
48	4	SHORT STRAP	STEEL NOTE 8	1-1/4 X .035 X 100
32	3	SLEEPER	WOOD NOTE 8	2 X [illegible] X 24
20	2	BACKUP CLEAT	WOOD NOTE 8	2 X 6 X 24
10	1	CROSSMEMBER	WOOD NOTE 8	2 X 6 X 84

LIST OF MATERIALS PAGE 3

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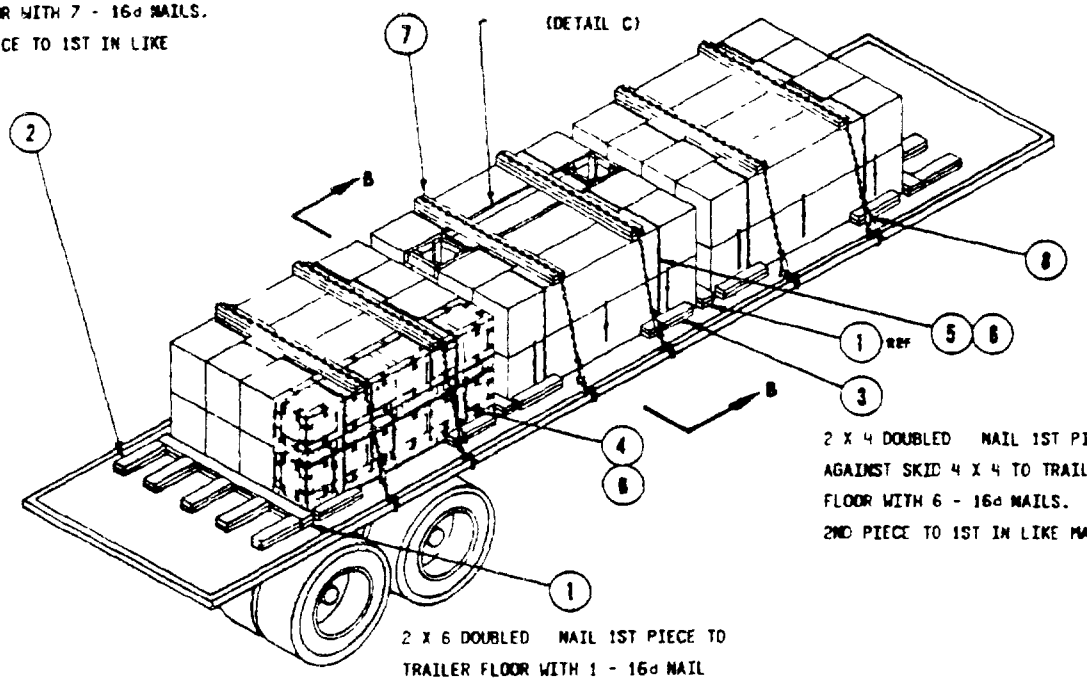
MIL-STD 1320 17SC (NAVY)

LTL 40 FT FLATBED TRAILER

NOTE
TARPAULIN OVER
CONTAINERS OMITTED
FOR CLARITY

2 X 6 DOUBLED NAIL 1ST PIECE TO TRAILER FLOOR WITH 7 - 16d NAILS. NAIL 2ND PIECE TO 1ST IN LIKE MANNER

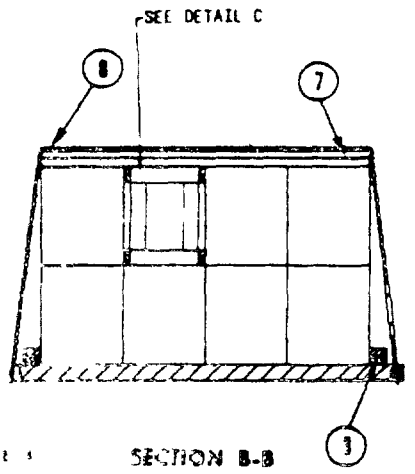
SEE FILLER ASSEMBLY (DETAIL C)



2 X 4 DOUBLED NAIL 1ST PIECE AGAINST SKID 4 X 4 TO TRAILER FLOOR WITH 6 - 16d NAILS. NAIL 2ND PIECE TO 1ST IN LIKE MANNER

2 X 6 DOUBLED NAIL 1ST PIECE TO TRAILER FLOOR WITH 1 - 16d NAIL EVERY 6 - 8 INCHES NAIL 2ND PIECE TO 1ST IN LIKE MANNER

LTL 40 FT FLATBED TRAILER



SECTION B-B

REQ'D	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
8	11	LATERAL SPACER	WOOD NOTE 8	2 X 4 X 17
8	10	VERTICAL SPACER	WOOD. NOTE 8	2 X 4 X 25
4	9	LONGITUDINAL SPACER	WOOD. NOTE 8	2 X 4 X 110
6	8	CHAIN & LOAD BINDER	NOTES 3, 4, 5 & 6	3/8 OR 5/16
12	7	PROTECTOR BOARD	WOOD. NOTE 8	2 X 6 X 80
56	6	SEAL	NOTE 8	1-1/4 X .035
24	5	LONG STRAP	STEEL. NOTE 8	1-1/4 X .035 X 150
32	4	SHORT STRAP	STEEL. NOTE 8	1-1/4 X .035 X 100
24	3	SLEEPER	WOOD NOTE 8	2 X 4 X 24
20	2	BACKUP CLEAT	WOOD. NOTE 8	2 X 6 X 24
8	1	CROSSMEMBER	WOOD. NOTE 8	2 X 6 X 86

LIST OF MATERIALS

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LTL 40, 44, 45 FT FLATBED TRAILERS

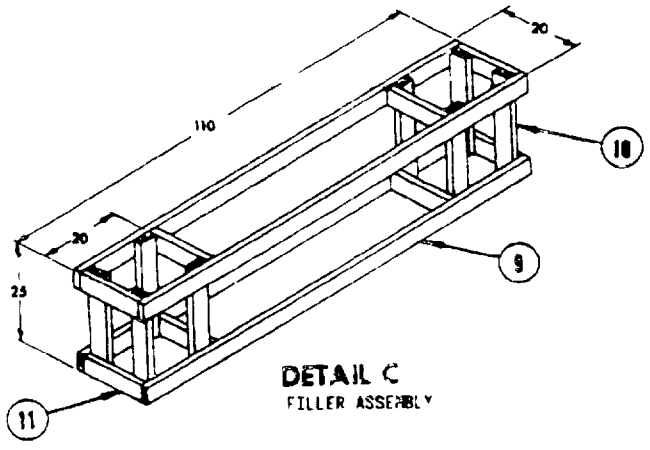
1 WHEN REQUIRED TO SHIP LESS-THAN-TRUCKLOAD SELECT THE CORRECT LOAD PATTERN FOR 40 44 AND 45 FT TRAILERS FROM TABLES II AND III. BLOCK AND BRACE USING PRINCIPLES SHOWN ON PAGES 3 4 AND 5

2 FOR STACKS 3 CONTAINERS WIDE USE CROSSMEMBERS 60 66 WIDE PROTECTOR BOARDS 60 WIDE AND 4 BACKUP CLEATS

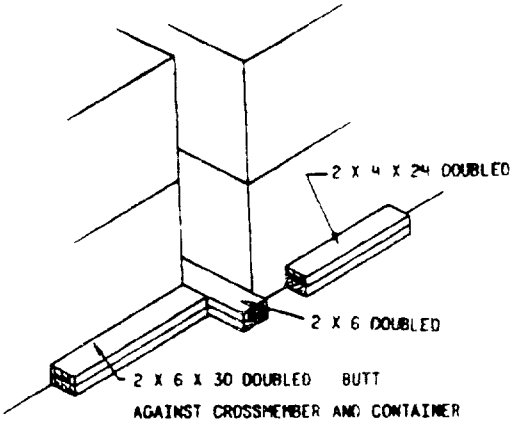
3 FOR STACKS 2 CONTAINERS WIDE, USE CROSSMEMBERS 40 - 48 WIDE PROTECTOR BOARDS 40 WIDE AND 3 BACKUP CLEATS

4 FOR SINGLE STACK, USE CROSSMEMBER 20 - 26 WIDE PROTECTOR BOARDS 20 WIDE AND 2 BACKUP CLEATS

5 WHERE 4 CONTAINER WIDE STACKS MEETS 3 CONTAINER WIDE AND 3 WIDE MEETS 2 WIDE, USE PRINCIPLE OF DETAIL D



DETAIL C
FILLER ASSEMBLY



DETAIL D

2 X 6 X 30 DOUBLED BUTT AGAINST CROSSMEMBER AND CONTAINER IN PLACE OF SLEEPER AND NAIL TO TRAILER FLOOR WITH 9 - 16d NAILS NAIL 2ND TO 1ST IN LIKE MANNER

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LTL FLATBED TRAILERS (CONTINUED)

TABLE II

LOAD PATTERN FOR 44 FT AND 45 FT
FLATBED TRAILERS

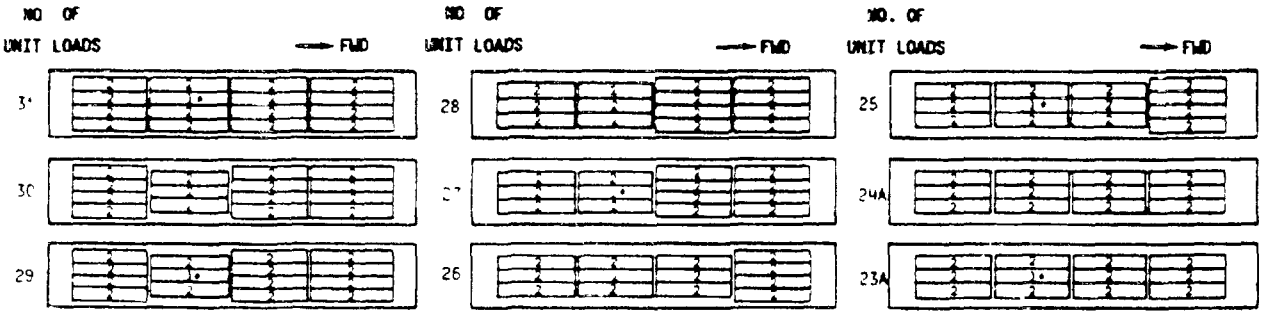
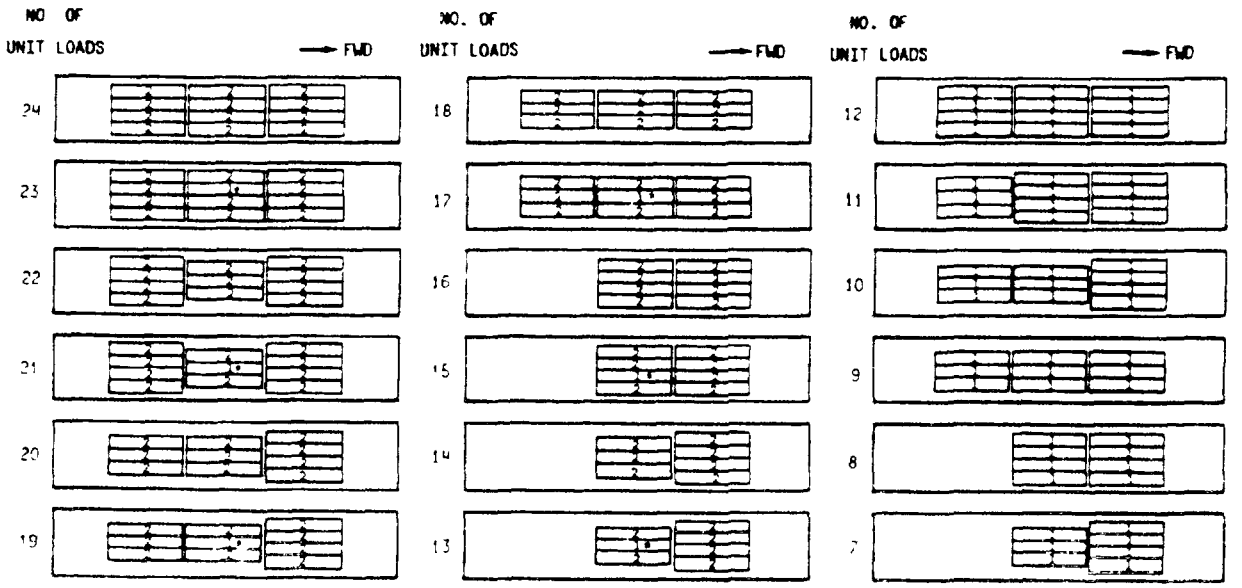


TABLE III

LOAD PATTERN FOR 40 FT 44 FT AND 45 FT
FLATBED TRAILERS

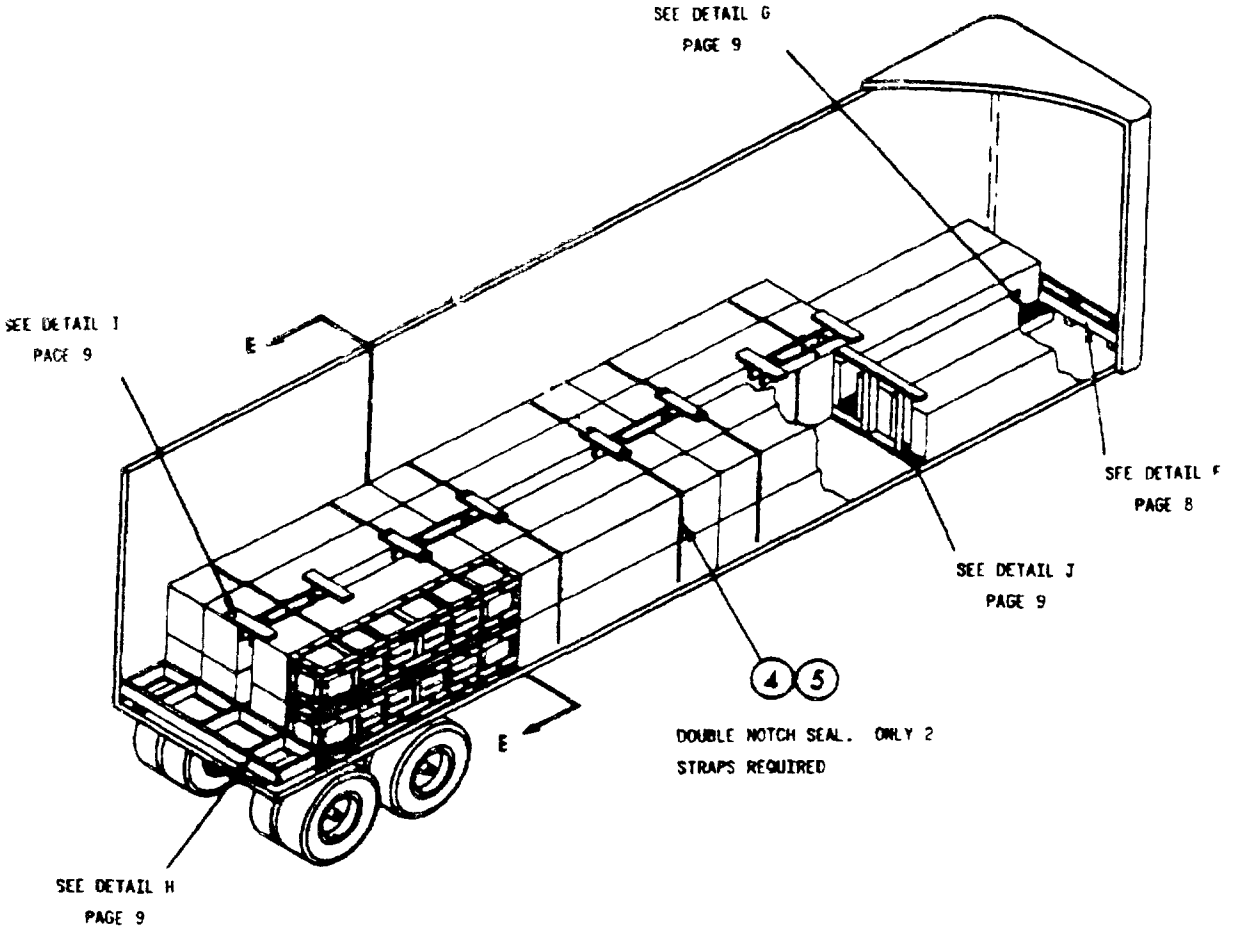


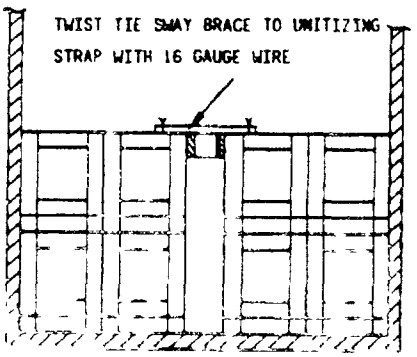
6 OR LESS. CONTINUE DECREASING NUMBER OF CONTAINERS AND STACKS OF CONTAINERS AS REQUIRED.

* REQUIRES FILLER ASSEMBLY, DETAIL C, PAGE 5, ON TOP AND UNIT(S) TO BOTTOM CONTAINER

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FTL 40 FT VAN (EMERGENCY SHIPMENTS ONLY)





SECTION E-E

*TW - TRAILER WIDTH

REQ'D	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
8	15	SUPPORT	WOOD, NOTE 8	2 X 6 X 24
8	14	KEEPER	WOOD, NOTE 8	2 X 6 TO SUIT
8	13	LONGITUDINAL	WOOD, NOTE 8	2 X 6 X 48
8	12	SPACER	WOOD, NOTE 8	2 X 6 X 5-1/2
6	11	LONG FILLER	WOOD, NOTE 8	2 X 6 X 55
4	10	UPRIGHT	WOOD, NOTE 8	4 X 4 X 36
2	9	CROSSPIECE	WOOD, NOTE 8	2 X 4 X TW*-1
4	8	RISER	WOOD, NOTE 8	2 X 4 TO SUIT
5	7	STRUT	WOOD, NOTE 8	2 X 6 TO SUIT
2	6	CROSSMEMBER	WOOD, NOTE 8	2 X 6 X TW*-1
12	5	SEAL	NOTE 8	1-1/4 X .035
24	4	UNITIZING STRAP	STEEL NOTE 8	1-1/4 X .035 X 150
1	3	FORWARD CROSSMEMBER	WOOD, NOTE 8	2 X 6 TO SUIT
4	2	VERTICAL POST	WOOD NOTE 8	4 X 4 X 11-1/2
1	1	AFT CROSSMEMBER	WOOD, NOTE 8	2 X 6 X TW*-1

LIST OF MATERIALS

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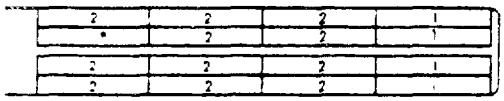
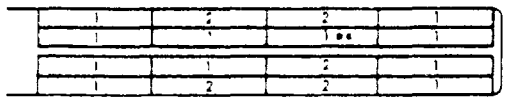
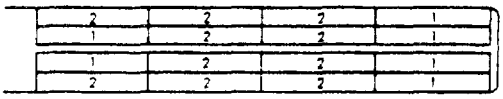
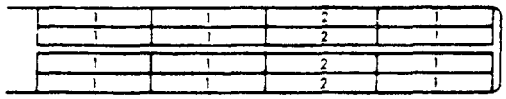
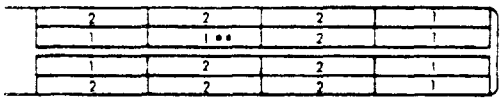
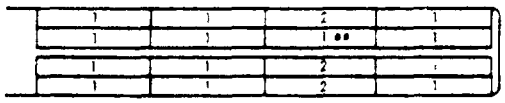
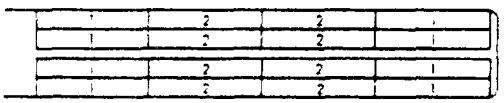
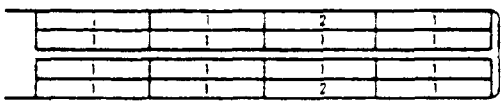
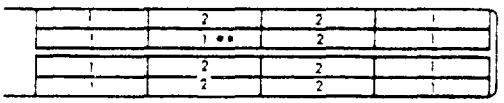
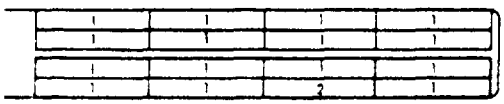
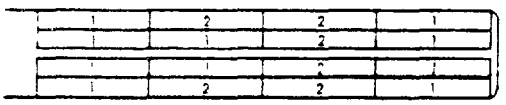
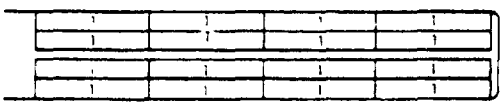
LTL 40 FT VAN

1 WHEN REQUIRED TO SHIP 27 OR LESS CONTAINERS IN 40 FT VAN SELECT THE CORRECT LOAD PATTERN FOR THE NUMBER TO BE SHIPPED FROM TABLE IV BLOCK AND BRACE USING THE PRINCIPLES SHOWN ON PAGE 7

2 WHEN USING FILLER ASSEMBLY, DETAIL C, STRAP THE ASSEMBLY ON TOP OF THE CONTAINER IN THE SAME MANNER AS TWO CONTAINERS ARE STRAPPED TOGETHER AS SHOWN IN THE FTL

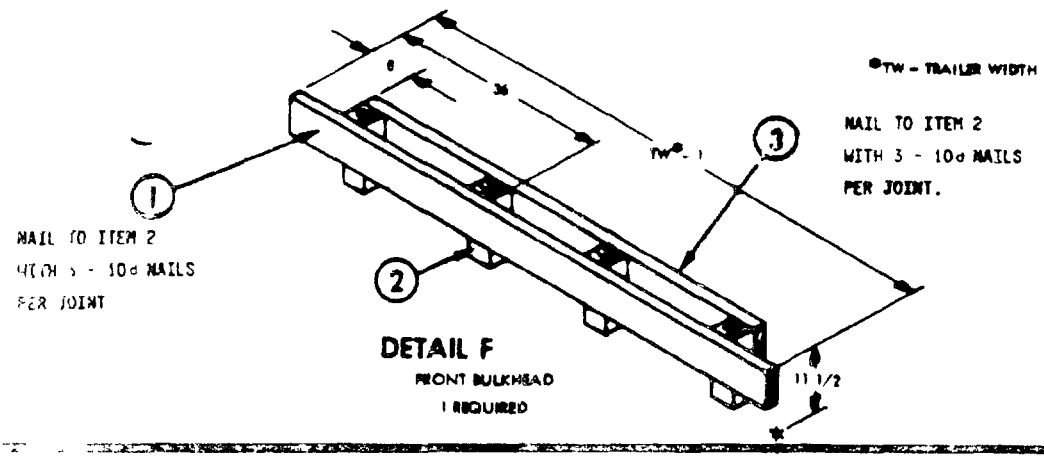
3 WHEN SHIPPING 12 TO 15 ROCKET MOTORS USE FILLER ASSEMBLIES TO FILL OUT LOAD PATTERN AS SHOWN FOR 16 MOTORS BLOCK AS SHOWN ON PAGE 7 WHEN SHIPPING 11 OR LESS UNIT LOADS THE TRAILER MUST HAVE A NAILABLE FLOOR EACH ROW OF CONTAINERS SHALL BE RESTRAINED IN THE AFT DIRECTION USING A DOUBLED CROSSMEMBER AND 2 DOUBLED 2 X 6 X 24 CLEATS NAILED TO THE VAN FLOOR WITH 7 - 10d NAILS PER LAYER

TABLE IV

NO. OF UNIT LOADS	LOAD PATTERN FOR VANS → FWD	NO. OF UNIT LOADS	LOAD PATTERN FOR VANS → FWD
27		21	
26		20	
25		19	
24		18	
23		17	
22		16	

15 OR LESS DO NOT USE TRAILERS WITH NON-NAILABLE FLOORS DECREASE THE NUMBER OF CONTAINERS IN THE LOAD PATTERN STARTING AT THE REAR OF THE LOAD BLOCK AND BRACE AS SPECIFIED IN LTL NOTE 3

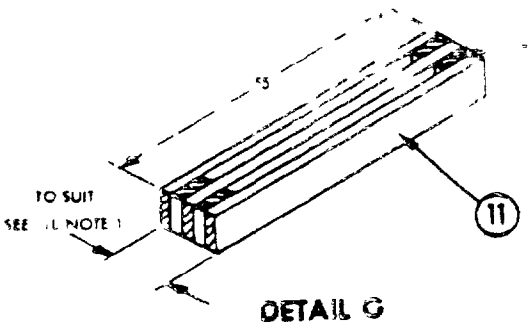
**REQUIRES FILLER ASSEMBLY, DETAIL C UNITIZED TO BOTTOM CONTAINER WITH TWO 1-1/4 X .035 STRAPS



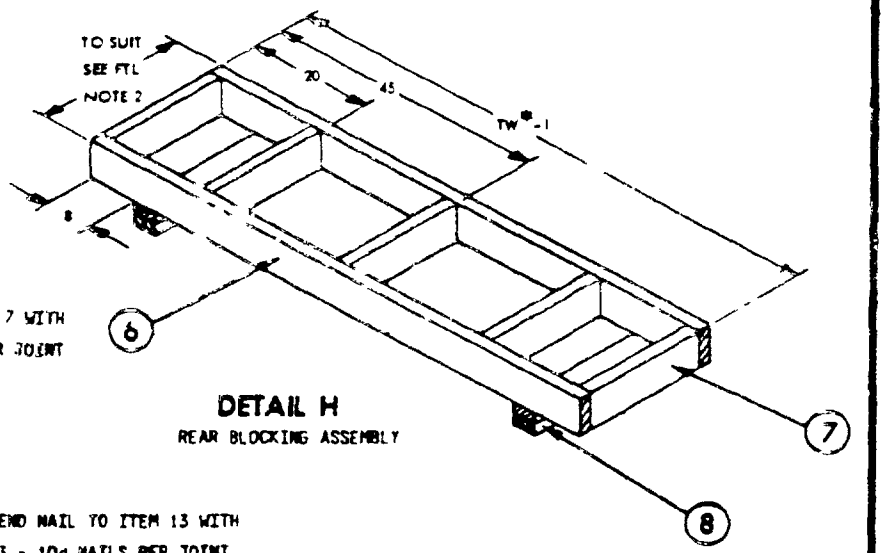
DETAIL F

FRONT BULKHEAD
1 REQUIRED

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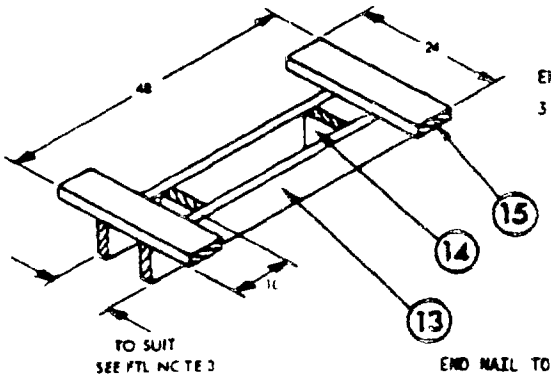
DETAIL G
FILLER



END NAIL TO ITEM 7 WITH
3 - 16d NAILS PER JOINT

DETAIL H
REAR BLOCKING ASSEMBLY

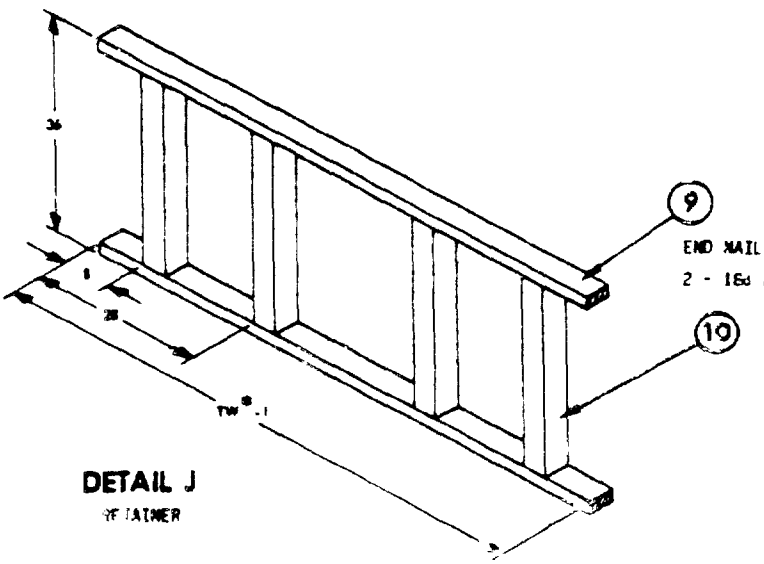
NAIL TOP SUPPORT TO ITEM 6 WITH 2 - 10d NAILS PER JOINT LAMINATE BOTTOM WITH 3 - 10d NAILS PER JOINT.



END NAIL TO ITEM 13 WITH
3 - 10d NAILS PER JOINT

END NAIL TO ITEM 14 WITH
3 - 10d NAILS PER JOINT.

DETAIL I
SWAY BRACE



END NAIL TO ITEM 10 WITH
2 - 16d NAILS PER JOINT

DETAIL J
RETAINER

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE: MIL-STD-1320-175C - Truckloading of Rkt Motor MK 27 or MK 56 and MODS (DURM) in Container, MK 593 Mod 0

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

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